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UKUDLA Data Science Certificate

Introduction to Data Science for Food Systems Analysis

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for Sustainable and Resilient Food Systems and Applied Agricultural and Food Data Science

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Why data science matters for food systems

Food systems generate diverse evidence

- laboratory measurements and assays;
- designed field experiments;
- field observations, surveys and qualitative evidence; and
- administrative, market, climate and remote-sensing data.

Decisions require more than data

- transparent questions and assumptions;
- appropriate preparation and analysis;
- reproducible computational workflows; and
- careful interpretation of uncertainty.

Data science connects questions, data, computation and judgment to produce evidence that can be inspected and reused.

The challenge is a chain of decisions

Question → Data → Analysis → Claim

- Available data may not represent the intended population or concept.
- Transformations and models encode assumptions that affect conclusions.
- Results can be technically correct but scientifically misleading.
- Documentation and reproducibility make the decision chain reviewable.

Data science is an integrated practice

Domain knowledge
Defines meaningful questions, measurements and interpretations.

Statistical reasoning
Describes variation, evaluates relationships and quantifies uncertainty.

Computing practice
Acquires, transforms, analyzes and communicates data reproducibly.

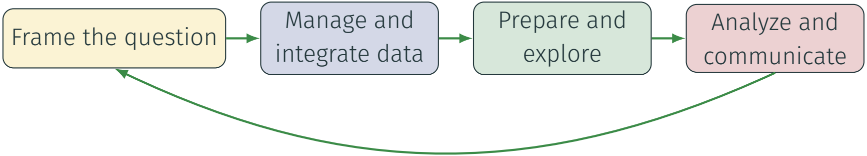
No single tool defines data science. Credible work depends on how these forms of knowledge are combined.

Different questions require different analyses

Purpose	Question	Example output
Descriptive	What patterns are present in the observed data?	Summaries, distributions and trends
Explanatory	How should an assumed relationship be estimated and interpreted?	Effect estimate with assumptions
Predictive	How accurately can an unknown outcome be predicted?	Out-of-sample predictions and error

- A useful analysis begins with a clearly stated purpose.
- Explanatory and predictive success are not interchangeable.
- Every conclusion is conditional on data quality and design choices.

A data science workflow is iterative



- Findings often reveal new data-quality issues or better questions.
- Iteration is expected; undocumented iteration is the risk.
- Versioned inputs, code and decisions preserve the path to a result.

Credibility requires reproducibility and responsibility

Reproducible practice

- version code and documentation;
- record software environments;
- automate transformations; and
- preserve provenance and validation.

Responsible practice

- respect licenses, privacy and consent;
- examine representation and bias;
- distinguish evidence from assumptions; and
- communicate limitations honestly.

A result is credible only when its production and interpretation can be examined.

Part 1 builds tools for reproducible work

Version control

Record meaningful project states and collaborate through Git and GitHub.

Environments

Recreate package and system dependencies with `renv` and Docker.

Remote computing

Work safely on other computers through SSH and Linux tools.

These tools support every later topic; they are part of the analytical method, not administrative extras.

Part 2 establishes trustworthy analytical data

Data management
Describe, organize, validate and preserve data and their history.

Data integration
Align identifiers, time, space and meaning across sources.

Data preparation
Transform managed data into an explicit unit of analysis.

Analytical validity begins before modeling: later results inherit the choices made in constructing the data.

Part 3 develops complementary forms of analysis

Topic	Primary role
Data visualization	Reveal structure and communicate evidence through graphical encodings
Descriptive analysis	Summarize distributions, dependence, change and stability
Explanatory analysis	Estimate and qualify relationships under causal assumptions
Predictive analysis	Evaluate forecasts on observations not used for model fitting

- Each topic combines motivation, reusable concepts and application guidance.
- Detailed pages support study; the example repository supports practice.

Key message: make the complete reasoning chain visible

Data science is the disciplined construction of evidence from questions, data, computation and assumptions—not merely the use of software or models.

- Start with the purpose of the analysis.
- Treat data and computational choices as part of the evidence.
- Match descriptive, explanatory or predictive methods to the question.
- Preserve enough information for others to inspect, reproduce and reuse the work.

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Version Control and Collaboration using Git & GitHub

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Why version a data science project?

```
analysis.R
analysis-new.R
analysis-final.R
analysis-final-revised.R
analysis-final-revised-2.R
```

Version control provides

- one authoritative project;
- a traceable sequence of changes;
- meaningful recovery points;
- parallel and reviewable work; and
- a link between decisions and results.

Saving a file is not the same as recording a version. Version control turns a folder of files into a project with a documented history.

Why this matters for food-systems data science

Food-systems analyses combine many changing elements:

- data-acquisition and cleaning decisions;
- definitions, units, study units, and time periods;
- statistical code and model specifications;
- figures, reports, and interpretation; and
- software-environment descriptions.

A small change in a filtering rule or unit conversion can change the scientific conclusion.
Git helps connect that conclusion to a specific, inspectable state of the project.

Git and GitHub: local history and shared collaboration

Git

- Distributed version control system
- Runs on the local computer
- Records project snapshots
- Most operations work offline

GitHub

- Online platform hosting Git repositories
- Provides a shared remote location
- Adds review and access control
- Supports issues, pull requests, and automation

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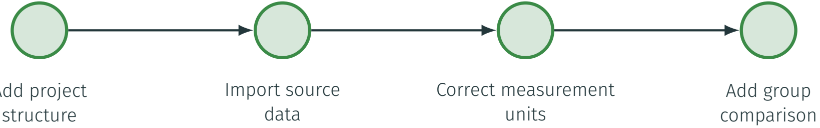
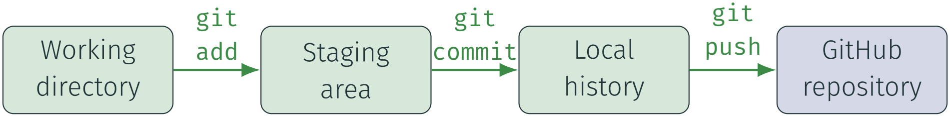
A commit is a meaningful project snapshot

A commit records selected changes with an author, timestamp, message, and connection to the preceding project history.



- A commit should contain one coherent change.
- The diff records *what* changed.
- The message should help explain *why*.

From an edited file to shared history



Workflow

- Editing changes the working directory.
- Staging selects what belongs to the next commit.
- Committing records a local snapshot.
- Pushing shares local commits with the remote repository.

Not everything should be tracked!

- secrets and credentials;
- confidential data;
- large downloaded data;
- generated outputs;
- package libraries and caches; and
- editor-specific state.

Inspect before you record

```
git status
git diff
git add path/to/file
git diff --staged
git commit -m "Correct measurement unit conversion"
git push
```

Questions before committing

- Do I understand each change?
- Do these changes belong together?
- Did I include generated or private files?

A useful commit

- is small and coherent;
- leaves the project usable; and
- explains why the change matters.

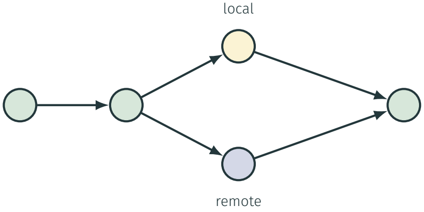
A collaborative Git cycle

`inspect` → `pull` → `edit` → `commit` → `push`

- Pull remote changes before beginning a new task.
- Make small, focused changes.
- Communicate which files or tasks people are working on.
- Push completed commits so collaborators can build on them.
- Integrate frequently; long-lived divergence increases effort.

Collaboration means exchanging and integrating commits—not emailing copies of "final" files.

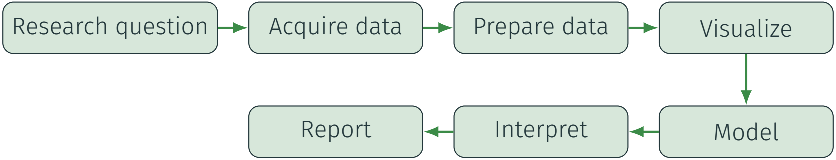
Divergence, merging and scientific judgment



- Diverging histories are normal.
- Git often combines different changes automatically.
- A conflict asks a human to choose or integrate.
- The integrated analysis must still run and retain its intended meaning.

Git identifies overlapping text. It cannot decide which research assumption, data transformation, or interpretation is scientifically correct.

Version control supports the complete data science workflow



Git and GitHub

- mark meaningful stages;
- connect code changes to results; and
- support sharing and review.

Track deliberately

- Track code, documentation, and lockfiles.
- Exclude secrets, sensitive data, caches, and unsuitable large files.

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Key message: version meaningful project states

1. Git records meaningful states of a project.
2. GitHub provides a shared location and collaboration tools.
3. Inspecting changes is part of responsible data science.
4. Small commits and frequent synchronization reduce problems.
5. Version control supports reproducibility, but does not replace data, environment, and workflow documentation.

Detailed setup, repository, workflow, and troubleshooting instructions are provided in the learning-module guides.

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Reproducible Environments using `renv` & Docker

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The reproducibility question

Can another learner

- run the analysis scripts?
- obtain the required packages?
- reproduce the same results?
- recreate the analysis next year?

Environments can differ

```
Learner A       Learner B
R 4.3.3         R 4.5.0
dplyr 1.1.4     dplyr 1.2.x
Ubuntu          Windows
```

Sharing code is necessary, but code alone does not describe the computational conditions under which it ran. “Works for me” is not yet a reproducibility strategy.

A computational environment has several layers

Project code, configuration and data

R packages and package versions

R version and publishing tools

System libraries and operating system

Computer infrastructure

Isolation
Keep project dependencies separate.

Portability
Move work between computers.

Reproducibility
Recreate recorded conditions.

What does `renv` manage?

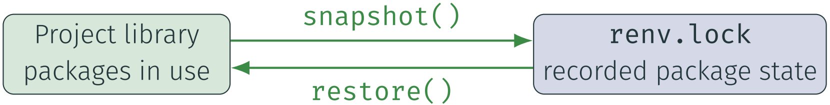
`renv` provides

- a project-local R package library;
- isolation from other projects;
- a record of package versions and sources; and
- tools to restore that state.

`renv` does not manage

- the complete operating system;
- all system libraries;
- hardware or external services; or
- the research workflow itself.

`renv.lock` is a machine-readable description of the project's R package environment and belongs in version control.



- `snapshot()` updates the lockfile from the project library.
- `restore()` recreates the project library from the lockfile.
- Review lockfile changes before committing them.
- A shared project can be initialized once and restored by collaborators.

What does a container add?

A container packages

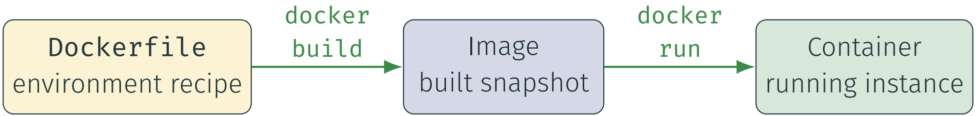
- an operating-system user space;
- system libraries and tools;
- R and publishing tools;
- project dependencies; and
- a default working context.

A container shares

- the host computer's kernel;
- host hardware and resources;
- explicitly mounted files; and
- configured network access.

Containers provide stronger environment isolation than a project package library, but are lighter than a complete virtual machine.

Dockerfile, image and container



- The Dockerfile describes how to construct an image.
- The image is an immutable snapshot that can be used repeatedly.
- Each container is a running instance created from that image.
- Rebuild after changing the Dockerfile or copied source code.

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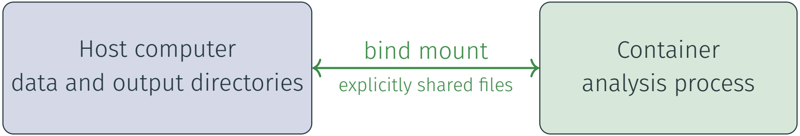
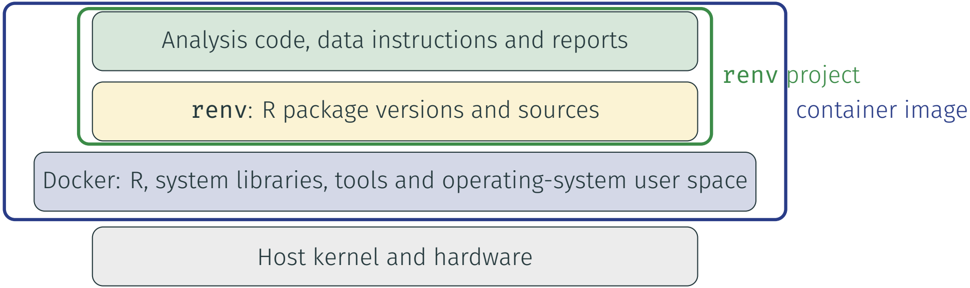
Containers are disposable; results must persist



- A container's writable layer normally disappears with the container.
- Bind mounts expose selected host directories inside the container.
- Inputs can be read and outputs written back to the host.
- Mount only what the analysis needs.

Two tools manage different layers



`renv` records the R package layer. Docker records a broader execution environment. Together they provide complementary reproducibility information.

Run one analysis reproducibly across contexts

Method	Environment	Typical use
Local batch	Host R with restored **`renv`** library	Run the complete workflow
Local interactive	RStudio or terminal R with **`renv`**	Develop and inspect
Docker batch	Image in a temporary container	Run a consistent environment
Docker interactive	Shell or R inside the container	Inspect the container context

Git repository → Execution environment → Analysis workflow → Data, figures, report

Key message: record every relevant environment layer

Environment tools can help recreate

- package versions;
- language and system tools;
- execution commands; and
- a consistent working context.

They cannot guarantee

- correct data or code;
- valid scientific assumptions;
- permanent external sources; or
- responsible interpretation.

`renv` records R packages; Docker records a broader environment. Together with Git, documented data, and a reproducible workflow, they make an analysis easier to recreate and inspect.

Detailed setup, repository, workflow, and troubleshooting instructions are provided in the learning-module guides.

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Remote Computing using SSH & Linux

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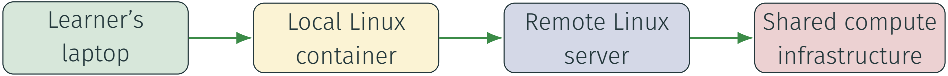
One project in different computing contexts



Linux

- Operating-system family
- Common on servers and containers

Terminal

- Text-based interface
- Displays input and output

Shell

- Reads and runs commands
- Bash is a common shell

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Why the command line matters in data science

Repeatable work

- Commands can be saved in scripts.
- The same instruction can process many files.
- Text commands are easy to document and review.

Portable work

- Remote systems may have no desktop.
- Small tools can be combined.
- Automation scales beyond manual clicking.

The command line is not only a different interface. It expresses repeatable operations in a form that computers and collaborators can execute.

SSH: one secure connection, different outcomes

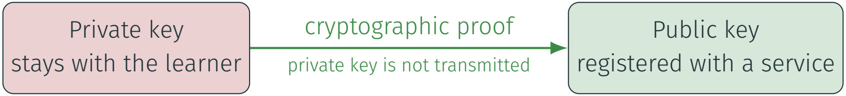
GitHub repository access

- Clone, fetch, pull, and push
- Exchange Git objects
- No general-purpose shell

Remote computing

- Open a remote shell
- Run programs and manage files
- Transfer files securely

Public-key authentication



- The public key may be shared; the private key must remain private.
- A passphrase protects the private key if its file is exposed.
- The server checks that the client controls the matching private key.
- Host fingerprints help verify that the client reached the intended host.

A remote session changes the computing context

Local context
local user
local files
local processes

SSH connection
keyboard and text cross the network

Remote context
remote user
remote files
remote processes

Always establish context before acting:

- Which computer and user am I controlling?
- Which directory and repository am I in?
- Which files and processes will this command affect?

Navigate and combine small Linux tools

```
/
|-- home/
|   `-- student/
|       `-- analysis-project/
|-- tmp/
`-- work/
```

```
head results/summary.csv |
  column -s, -t
```

Navigation and composition

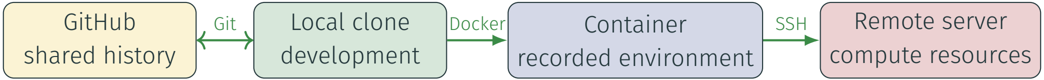
- Absolute paths begin at /.
- Relative paths begin at the current directory.
- A pipe connects one command's output to the next.

Commands operate in a context. Confirm the current directory before reading, writing, moving, or deleting files.

A container is a safe Linux practice environment

- The image supplies a consistent Linux environment.
- A temporary container can be explored and discarded.
- Source copied into the image is a build-time snapshot.
- Bind-mounted inputs and outputs remain on the host.

One data science project across several contexts



- Git identifies and exchanges project states.
- Docker describes and builds the execution environment.
- SSH provides secure access to another computer.
- Linux commands operate the project in containers and on servers.

Command-line and remote-computing care

Before a command

- Confirm the host and user.
- Confirm the current directory.
- Inspect targets and command options.
- Understand persistence and permissions.

For shared infrastructure

- Protect credentials and private keys.
- Respect CPU, memory, and storage limits.
- Manage long-running work appropriately.
- Preserve outputs and stop unneeded processes.

SSH secures the connection. It does not make every command safe, every server trusted, or every analysis scientifically correct.

Key message: know which computer a command affects

1. Linux, a terminal, a shell, and SSH are related but distinct.
2. SSH authenticates and protects communication with a remote service.
3. A remote shell operates another computer, not the local laptop.
4. Linux commands provide a repeatable interface across containers and servers.
5. Git, Docker, SSH, and Linux connect one data science project across several computing contexts.

Detailed commands, exercises, verification, and troubleshooting are provided in the learning-module guides.

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Reproducible analysis can reproduce a mistake

The workflow runs

- code is versioned;
- packages are restored;
- the container starts; and
- the report is rendered.

But the data may be misunderstood

- a measurement uses the wrong unit;
- missing values are read as zero;
- flags are ignored; or
- a revised source replaces an earlier file.

> Data management makes the meaning, origin, quality, organization, and permitted use of data explicit.

Data move through a lifecycle

Plan → Acquire → Describe → Validate → Use → Share → Preserve

- Decisions at one stage affect what is possible at later stages.
- Documentation and provenance connect the stages.
- Validation is repeated when data, requirements, or uses change.

This session moves from **motivation**, through the essential **concepts**, to their responsible **application** in a data workflow.

What does one row represent?

Dataset grain

The grain states what a single observation represents.

Depending on the study, one row may represent one:

- sample or specimen;
- plot or observational unit;
- measurement occasion;
- experimental treatment; or
- reported entity and period.

Role	Example
Identifier	Sample, plot, farm or region ID
Time	Collection date, season or year
Measure	Observed value
Meaning	Variable or assay
Context	Unit, treatment, method or flag

A key is the smallest set of variables expected to identify one row uniquely.

Food-systems evidence has different structures

Study structures

- Laboratory: aliquots within samples or batches
- Experiment: treatments assigned to field units
- Observation: farms, plots or people measured without assignment
- Secondary: reported entities and periods

Common dimensions

- Cross-sectional, longitudinal or repeated measures
- Hierarchical units, blocks, sites and batches
- Points, polygons, transects or raster cells
- Identifiers, protocols and boundaries may change

Structure determines valid keys, comparisons, summaries, and integration choices. A file format alone does not describe that structure.

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Metadata and provenance answer different questions

Source metadata

- definitions;
- methods;
- classifications;
- code lists; and
- quality flags.

Data dictionary

- variable names;
- meanings;
- types;
- units; and
- allowed values.

Provenance record

- collector or provider and dataset;
- protocol/version and collection/access date;
- license;
- retrieval method; and
- derived artifacts.

Metadata explain data; provenance explains their history.

Data quality means fitness for purpose

Completeness | Validity | Uniqueness | Consistency

Plausibility | Timeliness

- Quality requirements depend on the research question and intended use.
- Accuracy often cannot be established from a file alone.
- Field notes, laboratory flags and provider notes are part of the evidence.
- A quality concern should be reported before it is corrected or excluded.

Application: validate before transforming

Structural checks

- expected columns and types;
- row count and temporal coverage;
- uniqueness of the proposed key;
- duplicate records; and
- allowed codes and units.

Semantic checks

- missingness patterns;
- impossible or suspicious values;
- consistency with definitions;
- agreement with metadata; and
- unresolved quality flags.

Validation should reveal assumptions and anomalies. It should not silently “fix” the evidence.

Organize the project by data role

```
analysis-project/
|-- data-raw/
|-- data-interim/
|-- data-processed/
|-- metadata/
|-- scripts/
`-- reports/
```

- Preserve raw inputs unchanged.
- Create derived data with code.
- Keep dictionaries, licenses and provenance near the project.
- Use stable names, not `final-v2-revised.csv`.
- Decide deliberately which data belong in Git.

Raw, interim and processed describe roles in a workflow—not levels of correctness.

Manage the project data responsibly

Before storing or sharing

- Check consent, license and terms of use.
- Record citation and attribution.
- Confirm whether redistribution is permitted.
- Protect personal, confidential and location-sensitive data.

FAIR principles

- Findable
- Accessible
- Interoperable
- Reusable

FAIR does not necessarily mean open.

Farm, household, location and biological data can remain sensitive even after direct names are removed.

Key message: make data inspectable before analysis

Unchanged source data → Data dictionary and provenance → Validation script → Quality and limitations report

1. State the grain and expected key.
2. Interpret variables, units, codes and flags.
3. Document source, meaning, license and history.
4. Validate structure and meaning without changing the raw file.
5. Record unresolved concerns for the next workflow stage.

The next session asks how to obtain and integrate multiple sources.

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UKUDLA Data Science Certificate

Data Integration

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for Sustainable and Resilient Food Systems and Applied Agricultural and Food Data Science

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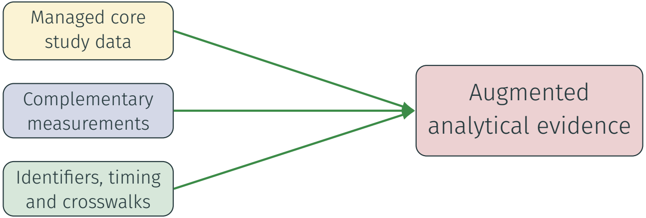
A managed dataset may not answer the question



One managed source rarely contains every outcome, exposure, treatment, covariate or context required by a study. Additional measurements or datasets can add evidence—if their observations can be aligned responsibly.

Data integration combines information from different sources into a coherent unit of analysis while preserving meaning and lineage.

Data integration extends data management

Manage each source
Describe, validate and preserve both the existing and additional data.

Integrate deliberately
Align identifiers, grains, periods, spaces, units and classifications.

Manage the result
Audit the join and document the new artifact, assumptions and lineage.

This session moves from **motivation**, through the essential **concepts**, to an auditable **application** across compatible sources.

Data acquisition supports this process; integration is the teaching focus.

Sources bring different representations

Core study data

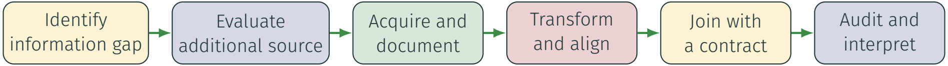
- laboratory samples or assay results;
- experimental plots and treatments;
- observational units and visits; or
- entities and periods from secondary data.

Complementary data

- instrument, sensor or laboratory measurements;
- management, survey or administrative records;
- weather, soil or remote-sensing products; and
- different grains may require aggregation.

Different representations must be made compatible without hiding how their meaning, space and time differ.

A data need initiates integration



The research question determines what information is missing, which source is appropriate, and what the integrated observation should represent.

Do not acquire an additional dataset before defining why and how it will be integrated.

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Compatibility has several dimensions

Structural compatibility

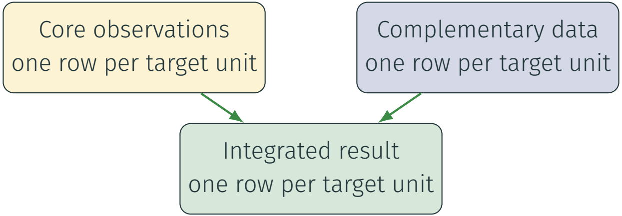
- identifiers and candidate keys;
- dataset grain and relationships;
- variable types and units;
- missing-value conventions; and
- classifications and code lists.

Scientific compatibility

- population and geographic support;
- observation and reporting periods;
- measurement definitions;
- aggregation assumptions; and
- fitness for the intended question.

A technically successful join does not prove that the combined variables refer to comparable evidence.

State the grain before joining



- A one-to-one join should preserve the expected key and row count.
- A one-to-many relationship can legitimately multiply rows.
- An unexpected many-to-many relationship often indicates an incomplete key.
- The join function cannot decide whether the scientific relationship is valid.

Stable identifiers and crosswalks connect sources

Names are fragile

- spelling and punctuation differ;
- abbreviations differ;
- names and boundaries change; and
- the same name can identify different entities.

Source label	Stable ID
Plot 1	SITE-A-P001
Sample 17	LAB-2026-0017
Farm North	FARM-0042

A crosswalk is a documented dataset, not an invisible correction embedded in code.

Check unmatched keys, duplicated keys, new missing values and row multiplication after every join.

Integration requires alignment, not only a join

Dimension	Possible mismatch	Decision to document
Identifiers	Sample, plot, farm and provider codes	Crosswalk and valid period
Time	Assay, visit, season or reporting period	Alignment and period definition
Space	Plot, site, region, point or raster cell	Boundary, CRS, extent and summary rule
Units	Mass, area, concentration, rate or index	Conversion and dimensional consistency
Schema	Names, types, categories and flags	Mapping and retained information

Aggregating replicates, visits, sensor readings or gridded values is an analytical choice, not a neutral format conversion.

Validate the result and preserve lineage

Join audit

- expected and actual row counts;
- matched and unmatched keys;
- duplicated or multiplied records;
- missingness introduced; and
- resulting grain and coverage.

Lineage record

- source of every retained variable;
- crosswalks and conversion rules;
- temporal or spatial aggregation;
- script and input versions; and
- known losses and limitations.

A critical failed expectation should stop the pipeline visibly rather than produce a plausible-looking result silently.

Key message: integration requires explicit alignment

Managed core dataset → Relevant additional dataset → Align and join stable keys → Audit and document integrated data

1. Identify the additional information required by the question.
2. Reconcile grain, identifiers, time, space, definitions and units.
3. Map source labels to stable analytical identifiers.
4. Join on tested keys under an explicit integration contract.
5. Save the derived table, audit, dictionary and lineage record.

A successful join is not sufficient: integrated data inherit every source and alignment limitation.

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UKUDLA Data Science Certificate

Data Preparation

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Integrated data are not automatically analysis-ready

The sources are managed and joined

- inputs are identified and documented;
- keys and coverage are validated;
- identifiers, space and time are aligned; and
- the join is audited.

The analysis still needs decisions

- which observations are retained?
- which shape should the table have?
- which units and types are required?
- which variables should be derived?

Data preparation turns managed evidence into a documented representation suited to a defined analysis.

Preparation choices affect the result

Managed and integrated evidence → Filter, reshape, convert, derive → Analysis-ready observations → Visualizations, summaries, models

- Dropping a record changes the population represented.
- Aggregation hides variation and requires a summary rule.
- Unit conversion changes values but should preserve meaning.
- Derived variables encode assumptions that later results inherit.

This session moves from **motivation**, through the essential **concepts**, to a reproducible **application** for an analytical dataset.

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Related activities answer different questions

Activity	Central question	Typical evidence
Validation	Do data meet documented expectations?	Checks and findings
Cleaning	Is a value demonstrably erroneous, and how is it corrected?	Justified correction record
Integration	How are compatible observations connected across sources?	Join contract and audit
Preparation	How are data represented for the intended analysis?	Reproducible transformations

A surprising value is not automatically an error. Preserve the evidence, investigate its meaning, and separate correction from transformation.

Begin with the target observation

Analysis contract

Before transforming, state:

- intended population;
- observation grain;
- candidate key;
- outcome and other variables;
- units and time periods; and
- missing-value policy.

Example analysis target

One row represents one region and year, containing an outcome and explanatory measures aligned to that period.

Candidate key

`region_id + year`

The target grain determines which transformations are valid.

Reshape structure without hiding meaning

Long source representation

Unit	Measure	Value
Plot 12	Biomass	2.34
Plot 12	Soil moisture	18.6
Plot 12	Nitrogen	40

Wide analytical representation

Unit	Biomass	Soil moisture
Plot 12	2.34	18.6

One target unit becomes one row with a separate variable for each retained measure.

Reshaping is valid only after the source key is known and every output cell is expected to receive at most one value.

Make every transformation explicit

Operation	Question	Evidence to preserve
Select/filter	Which population and variables remain?	Inclusion rule and row counts
Parse/recode	How are types, dates, labels or codes interpreted?	Mapping and failed parses
Convert	Are definitions and dimensions compatible?	Formula, source and target units
Reshape	What becomes an observation or variable?	Input and output grain and keys
Handle missingness	Is absence unknown, structural or introduced?	Missingness before and after
Derive	What assumption defines the new variable?	Formula, inputs and valid domain

Do not silently delete duplicates, replace missing values, cap outliers, or coerce failed values merely to make code complete.

Preparation can occur around integration

Managed source input → Source-specific preparation → Integrate additional data → Final analytical preparation

- Source-specific preparation makes source grains compatible.
- Integration connects compatible observations across sources.
- Final preparation selects or derives variables for a stated analysis.
- Each stage writes a new artifact; no stage edits a managed input.

Workflow order is determined by dependencies, not by topic boundaries.

Prepare an analysis-specific dataset

Validate before reshaping

- expected variable–unit pairs;
- unique unit–occasion–measure keys;
- known unit, treatment and period coverage; and
- numeric values and missingness.

Transform deliberately

- retain observations relevant to the question;
- reshape measures into analytical variables;
- harmonize units under documented formulas;
- derive transformations only on valid domains; and
- arrange one row per target observation.

The conversion formula, log domain, output grain, row count and candidate key are part of the preparation contract.

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Audit preparation and preserve lineage

Preparation audit

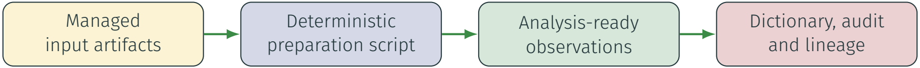
- input and output rows;
- unique input and output keys;
- failed parses or mappings;
- missingness before and after;
- valid units and ranges; and
- unchanged managed inputs.

Lineage record

- input and output artifacts;
- script and relevant parameters;
- filters and reshaping rules;
- conversion and derivation formulas;
- information lost; and
- dictionary updates.

Transformations learned from observed distributions—such as imputation or scaling—must later be estimated from training data only to prevent leakage.

Key message: analysis-ready is purpose-specific



1. State the target population, grain, key, variables and units.
2. Validate each assumption before applying its transformation.
3. Write derived artifacts instead of editing managed inputs.
4. Check row counts, keys, missingness, units and valid domains.
5. Document how every output variable was created.

Analysis-ready means ready for a stated purpose—not universally clean or final.

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UKUDLA Data Science Certificate

Data Visualization

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Prepared data do not reveal their patterns by themselves

A table preserves exact values

- study units and occasions are explicit;
- units and variables are documented;
- observations can be queried; and
- computations are reproducible.

A visual representation can reveal

- distributions and unusual values;
- differences between groups or units;
- change over time; and
- relationships between variables.

Data visualization maps documented observations to visual properties so that patterns can be inspected and communicated.

A chart is an analytical argument

Question and audience → Selection, encoding, scale and grouping → Visible pattern and comparison → Interpretation and limitations

- Exploratory graphics support investigation and iteration.
- Explanatory graphics focus attention on a defensible message.
- Both depend on choices about inclusion, aggregation and visual encoding.
- Neither turns an association into a causal effect.

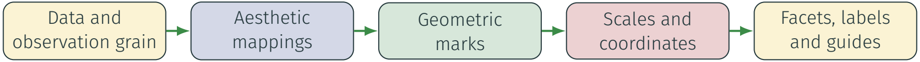
This session moves from **motivation**, through essential **concepts**, to a reproducible **application** in an analytical workflow.

Start with the analytical question

Question	Relevant structure	Possible graphic
How are values distributed?	One quantitative variable	Histogram, density plot, boxplot
How do groups compare?	Quantitative value and category	Aligned points, boxplots, faceting
How does a measure change?	Quantitative value ordered by time	Line or point plot
How do two measures vary together?	Two quantitative variables	Scatterplot
Where does a measure occur?	Measure and valid spatial geometry	Map

Choose a graphic from the question and variable types—not from a menu of decorative chart forms.

Build a graphic from explicit components



- Map variables to position, colour, shape, size or grouping.
- Select marks—points, lines, bars or areas—that support comparison.
- Use faceting when small multiples clarify group differences.
- Add labels and annotations that make definitions and units visible.

In `ggplot2`, every visible element should follow from an explicit layer or mapping.

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Use the strongest encodings for important comparisons

Usually easier to compare

1. position on a common scale;
2. position on aligned scales;
3. length; and
4. angle or area.

Use with care

- colour hue distinguishes categories;
- colour lightness can show order;
- size is read through area, not radius;
- shape supports few categories; and
- redundant cues improve accessibility.

Encode the quantity that matters most with position when possible; reserve colour and size for distinctions they can carry reliably.

Scales and aggregation change what the viewer sees

Choice	Risk	Review question
Axis limits	Differences appear exaggerated or observations disappear	Does the scale support an honest comparison?
Transformation	Distances acquire a different interpretation	Is a log or other scale named and justified?
Aggregation	Within-group variation becomes invisible	What grain and summary rule produced each mark?
Binning	Distribution shape changes with break placement	Are conclusions stable across reasonable bins?
Overplotting	Dense observations conceal frequency	Would transparency, jitter, bins or facets help?
Free facet scales	Shapes are comparable but magnitudes are not	Does each panel need the same quantitative scale?

Never hide a consequential filtering, transformation or aggregation decision in plotting code.

Design for interpretation, access and reuse

Make meaning visible

- informative title and subtitle;
- axis labels with units;
- readable legends or direct labels;
- source and aggregation note; and
- annotation of the intended comparison.

Make the figure usable

- sufficient contrast and text size;
- colour-blind-safe palettes;
- meaning not carried by colour alone;
- appropriate dimensions and file type; and
- reproducible code and stable output path.

A self-contained figure states what is shown, how it was measured and what it cannot establish.

Ask complementary questions of prepared data

One outcome across a study

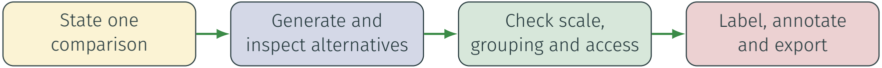
- distribution of the outcome;
- differences between treatments, sites or groups;
- change over time or measurement order; and
- unusual units or measurement occasions.

Outcome and a second variable

- distribution and patterns of the second variable;
- outcome versus predictor;
- differences across relevant groups; and
- possible non-linearity or overplotting.

Each chart should state its observation grain: one mark may represent a sample, plot, visit, reported entity, group summary or aggregated bin.

Turn an exploratory plot into a communication graphic



- Begin with a trend, distribution or relationship plot tied to one question.
- Compare faceting, grouping and scale choices against the intended question.
- Remove decoration that does not carry information.
- Add units, study context, alignment rules and a non-causal interpretation.
- Save the figure through code rather than manual export.

Key message: every visual choice shapes interpretation

Reproducible artifacts

- plotting script;
- exploratory figure set;
- one communication-ready figure;
- documented output paths; and
- source data and package environment.

Review evidence

- question matches the graphic;
- grain, units and scales are visible;
- grouping and aggregation are justified;
- design remains accessible; and
- interpretation respects limitations.

Visualization reveals and communicates patterns; the next session quantifies distributions and associations with descriptive statistics.

Show the evidence clearly—and make every consequential visual choice inspectable.

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Descriptive Analysis

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Visual patterns need numerical descriptions

Visualization helps us see

- distributions and unusual values;
- differences between treatments, sites or groups;
- changes over time; and
- possible associations.

Description helps us state

- how many observations contribute;
- what is typical and how values vary;
- how groups and periods differ; and
- how strongly variables vary together.

Descriptive statistics compress observed data into inspectable quantities; they complement rather than replace plots and individual observations.

A summary depends on population, group and time

Question and comparison → Population, grain, group and period → Count, location, dispersion, association → Interpretation and limitations

- The same mean can conceal different variation or skewness.
- A pooled summary can conceal group, batch, site or period differences.
- Missing values change the effective number of observations.
- A relationship observed in one period may not persist in another.

This session moves from **motivation**, through essential **concepts**, to a reproducible **application** for analytical data.

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Begin with the observations being described

State the scope

- target population and observed sample;
- observation grain and candidate key;
- categorical and numerical variables;
- groups, periods and weighting; and
- missing-value rule.

Report the denominator

- total rows;
- non-missing observations;
- counts by relevant group and period;
- frequencies and proportions; and
- observations excluded from a statistic.

A statistic without its population, grouping and effective sample size is incomplete.

Describe location, dispersion and shape together

Feature	Measures	Interpretation and caution
Location	Mean, median, quantiles	Mean uses every value; median is less sensitive to extremes
Dispersion	Range, IQR, variance, standard deviation	Range depends on extremes; SD shares the variable's unit
Shape	Skewness, tails, modes, unusual values	Needs visual inspection; an unusual value is not automatically an error
Coverage	Count, missing count, proportion available	The effective sample can differ between variables and groups

Use a small set of complementary summaries. A single average cannot represent distribution, variability, coverage and shape.

Association measures quantify co-variation

Covariance

- sign indicates joint direction;
- magnitude depends on both units;
- sensitive to extreme observations; and
- difficult to compare across scales.

Pearson correlation

- standardizes linear co-variation;
- ranges from -1 to 1;
- can hide nonlinear patterns; and
- changes with groups, periods and outliers.

Always pair association measures with a plot and the contributing observation count. Correlation is neither explanation nor causation.

Stationarity asks whether the process remains stable

Descriptive intuition

Across time, ask whether:

- the distribution shifts;
- the typical level changes;
- variability changes; and
- associations change.

Weak stationarity

A time process has:

- a constant mean;
- a constant variance; and
- covariance depending on lag, not calendar time.

Stationarity is an assumption about a data-generating process. A finite dataset can provide evidence for or against it, but cannot prove it universally.

Look for evidence of change before modeling

Pattern	Descriptive evidence	Why later models care
Trend	Period means or medians differ systematically	A constant-mean model is inappropriate
Changing variance	Period SDs or IQRs differ	Errors and uncertainty may not be stable
Structural change	An abrupt level or relationship shift	Historical parameters may not transfer
Seasonality/cycles	Repeated time-linked patterns	Time dependence remains in observations
Association drift	Correlations differ by group, site or period	One pooled slope may conceal changing relationships

Compare plots and grouped summaries across meaningful periods. Formal stationarity tests are optional extensions, not automatic verdicts.

Describe an outcome by group and period

For every group, report

- total and non-missing *n*;
- mean and median outcome;
- quartiles and IQR;
- standard deviation and range; and
- first and last occasion represented.

Compare periods

- an earlier historical period;
- a recent development period;
- a later evaluation period;
- levels, dispersion and coverage; and
- agreement with trend graphics.

Period summaries make potential distribution shift visible before a model is fit.

Describe relationships without pooling blindly

Overall summary and scatterplot → Group-specific summaries → Period-specific summaries → Agreement, drift and limitations

- Report each variable's location, dispersion, range and missingness.
- Calculate covariance and correlation with their contributing *n*.
- Compare pooled, group-specific and period-specific associations.
- Investigate whether between-group differences drive the pooled result.
- Retain measurement, aggregation and non-causal interpretation boundaries.

Key message: describe stability before modeling

Reproducible artifacts

- descriptive-analysis script;
- group and period summary tables;
- association table with effective *n*;
- compact report with linked figures; and
- explicit grouping and missingness rules.

Modeling hand-off

- evidence of trends or level shifts;
- evidence of changing variability;
- stability of associations;
- implications for time-based splitting; and
- questions requiring a stated model.

Description does not fix non-stationarity. It reveals when later explanatory and predictive models must represent time, change or limited transferability.

Summarize what was observed—and expose what may not remain stable.

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Explanatory Analysis

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Association is not yet a causal explanation

Descriptive question

- Do an exposure and outcome vary together?
- How does the association differ by group, site or period?
- Which observations support the pattern?

Causal question

- How would the outcome change under another intervention?
- Compared with which treatment or exposure condition?
- For which target units, settings and period?

An observed correlation compares different observations. A causal effect compares potential outcomes for the same target units under alternative exposures.

A model can quantify an association; assumptions give it causal meaning.

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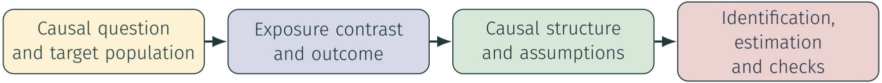
Begin causal analysis before fitting a model



- Define what intervention or exposure difference the effect represents.
- State which variables occur before, during and after the exposure.
- Use domain knowledge to identify common causes and selection processes.
- Decide whether the available data can identify the intended effect.

Causal analysis is a research design supported by models—not a regression command.

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Define the estimand before the estimator

Target quantity

- target population;
- exposure and contrast;
- outcome and time horizon; and
- aggregation level.

Potential outcomes

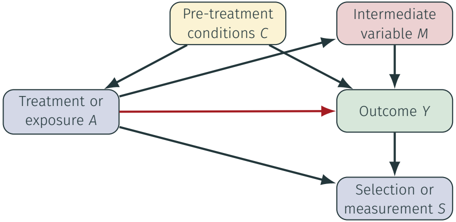
- $Y_i(a)$: outcome under condition a;
- individual effect: $Y_i(a_1) - Y_i(a_0)$; and
- average effect over target units.

Fundamental problem

- one unit experiences one realized exposure;
- the alternative outcome is unobserved; and
- identification requires valid comparisons.

> "Apply treatment A" must specify its dose, timing, delivery, comparator and target unit. Otherwise the causal contrast is ambiguous.

A causal diagram makes assumptions inspectable



Use the diagram to ask

- Which variables cause both exposure and outcome?
- Which paths should adjustment block?
- Is a variable a mediator or collider?
- Which causes are measured adequately?

Arrows encode assumptions, not correlations discovered by software.

Regression estimates a conditional association

$$Y_i = \beta_0 + \beta_1 A_i + \boldsymbol{\gamma}^\top \mathbf{C}_i + \varepsilon_i$$

Model components

- Y_i: outcome for target unit i;
- A_i: treatment or exposure condition;
- $\mathbf{C}_i$: selected pre-exposure covariates;
- $\boldsymbol{\gamma}$: their conditional coefficients; and
- ε_i: unexplained model deviation.

Coefficient interpretation

- β_1: modeled outcome contrast per exposure contrast;
- conditional on included pre-exposure covariates;
- linear and constant unless the model says otherwise;
- causal only when identification assumptions hold.

Adding controls changes the comparison; it does not automatically remove bias.

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Identification requires assumptions beyond regression

Assumption	Meaning	Questions across study designs
Consistency	Observed outcome matches the specified exposure condition	Are protocol, dose and timing well defined?
Exchangeability	Treatment groups are comparable conditional on design or covariates	Was treatment randomized; were common causes measured?
Positivity	Relevant contrasts occur for all adjustment profiles	Does every block, site or subgroup support the contrast?
No interference	One unit's treatment does not change another unit's outcome	Can plots, samples, farms or regions affect each other?
Measurement	Exposure, outcome and confounders represent intended concepts	Are instruments, assays and reported variables valid?

Identification connects the causal estimand to the observed-data distribution. If essential [illegible]ns are implausible, a precise estimate can remain causally uninterpretable.

Diagnostics assess the model, not the causal design

Model checks can reveal

- nonlinear exposure-outcome patterns;
- changing residual variance;
- influential observations;
- batch, block, site, temporal or grouped residual structure; and
- sensitivity to transformations and specification.

They cannot establish

- absence of unmeasured confounding;
- a well-defined intervention;
- correct causal direction;
- adequate measurement; or
- transferability beyond the target data.

Confidence intervals quantify sampling uncertainty under the model. They do not include unknown confounding, measurement error or misspecified causal structure.

Specify a causal analysis before estimation

Provisional target

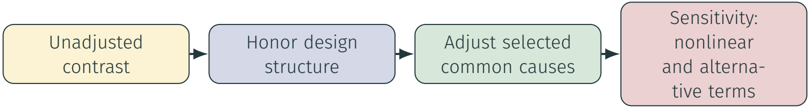
- units and target population;
- a well-defined exposure or intervention;
- outcome and timing;
- a meaningful exposure contrast; and
- target period and setting.

Required design work

- draw and justify a causal diagram;
- distinguish pre-exposure confounders from mediators;
- evaluate assignment or overlap within relevant groups;
- document unmeasured causes and exposure ambiguity; and
- decide whether the effect is identifiable.

Observed exposure variation is not automatically equivalent to a controlled intervention.

Compare estimates—then challenge their interpretation



- Report the target coefficient or contrast, uncertainty and effective *n*.
- Explain how the comparison changes across specifications.
- Inspect exposure overlap, residual plots, influence and temporal structure.
- Separate statistical precision from identification credibility.
- Label estimates as adjusted associations when causal assumptions remain unsupported.

Specification sensitivity is evidence to interpret—not a contest to find significance.

Key message: causal meaning comes from design

Reproducible artifacts

- causal question and estimand;
- causal diagram with justification;
- model specification table;
- coefficient and uncertainty table;
- diagnostics and sensitivity results; and
- code-linked explanatory report.

Conclusion must distinguish

- what the data describe;
- what the models estimate;
- which assumptions support identification;
- which assumptions remain doubtful; and
- which additional data or design are needed.

A credible causal analysis may conclude that the available data identify only adjusted associations. Making that boundary explicit is a scientific result, not a failure.

Estimate transparently—and earn the causal interpretation through design and assumptions.

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Explanation and prediction answer different questions

Explanatory objective

- What causal relationship is of interest?
- Which assumptions identify the effect?
- How should a coefficient be interpreted?
- Which alternative explanations remain?

Predictive objective

- What outcome must be predicted?
- Which information is available at prediction time?
- How accurate are predictions on unseen observations?
- Where and when does performance fail?

A variable can improve prediction without causing the outcome. A causally meaningful model can still predict poorly when the future differs from its training data.

Choose the objective before choosing the model or metric.

Evaluation must reproduce the intended use

Intended use and decision time → Target, unit, horizon and information set → Training and unseen test data → Errors, subgroups and limitations

- Training performance measures how well a model fits data it has already seen.
- Test performance estimates generalization only when the test mimics future use.
- Time-dependent data usually require earlier training and later testing.
- Any information unavailable at prediction time creates leakage.

A convenient random split can answer the wrong performance question.

Define a prediction contract

Prediction target

- outcome and unit;
- target population;
- prediction horizon; and
- scale and required output.

Information set

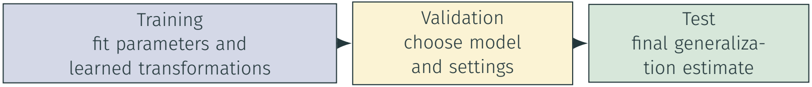
- prediction time;
- variables available then;
- data latency and revisions; and
- allowed history.

Evaluation contract

- split strategy;
- baselines and candidates;
- metrics and subgroups; and
- intended and excluded uses.

Predicting before sampling, during an experiment, or after data collection are different tasks because their horizons and available information differ.

Training, validation and testing have distinct roles



- Fit imputation, scaling, feature selection and model parameters on training data only.
- Use validation or resampling to compare alternatives without consuming the test set.
- Inspect the test set once the workflow is fixed; repeated tuning turns it into validation data.
- Preserve groups and temporal order when they define the deployment problem.

The test set protects an honest question only while it remains unseen.

Leakage makes performance look better than use

Common leakage routes

- future outcomes or revised future records;
- preprocessing learned from all observations;
- selecting variables after test inspection;
- duplicate or related units across splits;
- predictors measured after the prediction time.

Prevent leakage through design

- timestamp every input by availability;
- split before learned preprocessing;
- fit the complete recipe on training data;
- preserve group and time boundaries;
- document every test-set interaction.

A measurement can be a predictor only if it is available when the intended prediction is made. Otherwise the task and its claimed horizon must be redefined.

Baselines define the minimum useful performance

Candidate	Prediction rule	What it tests
Historical mean	Predict one training-period average	Can any structure beat a constant?
Group mean	Predict each relevant group's training average	Is group identity enough?
Linear trend	Extrapolate a training-period time trend	Does temporal structure help?
Group + trend	Combine group levels and time	Does structured regression generalize?
Added predictors	Use variables available at prediction time	Does extra complexity earn its cost?

- A baseline must follow the same split and information rules as every candidate.
- More complex models must improve relevant unseen-data performance, not only training fit.
- ity, stability, computation and interpretability remain deployment considerations.

Metrics summarize errors—and hide structure

Prediction error

$$e_i = y_i - \hat{y}_i$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |e_i|$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}$$

Interpret with context

- MAE weights absolute errors linearly;
- RMSE emphasizes larger errors;
- units follow the modeled outcome scale;
- averages can hide group, site or period failures;
- uncertainty grows with small test sets.

Generalization depends on similarity between evaluation and use. Batch effects, site differences, temporal shifts and new populations can make test performance optimistic.

Define a prediction task before model fitting

Prediction contract

- target, unit and prediction scale;
- target population and observation grain;
- prediction time and horizon;
- development and evaluation periods;
- information available at prediction time.

Boundary

- test outcomes remain unseen during fitting;
- the evaluation represents the intended population;
- temporal prediction may require extrapolation;
- unavailable future information is excluded;
- supported and excluded uses are documented.

The split must ask whether patterns learned from available data transfer to the intended use.

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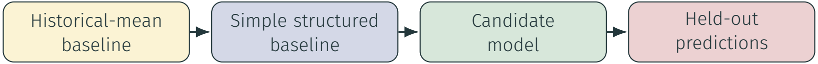
Fit on development data and predict held-out observations



- Learn every transformation and parameter from development data only.
- Generate one prediction from each candidate for every held-out observation.
- Preserve observed outcomes only for evaluation after prediction.
- Save model objects, split definitions and row-level predictions reproducibly.
- Add predictors only when their availability matches the prediction time.

Key message: evaluate the complete predictive workflow

Evidence to deliver

- prediction contract and split manifest;
- row-level held-out predictions;
- MAE and RMSE overall and by relevant group;
- error plots over groups and time;
- comparison with every baseline; and
- reproducible model artifacts and report.

Limitations to state

- small test sets provide limited evidence;
- performance can differ across groups;
- future distribution shifts remain possible;
- log-scale errors need outcome-scale context;
- predictive accuracy does not imply causality.

A predictive model is useful only for a stated task when it improves on relevant baselines, survives realistic unseen-data evaluation and remains reproducible at the point of use.

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Report where the model works, where it fails and where the evidence ends.

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